

Principles of Computer Vision, Lecture 1

Darius Burschka

<http://www6.in.tum.de/burschka/courses/vision>

Recordings and Slides on
<https://mvp.in.tum.de/courses/vision>

Slide access login: vision
password: vision_slides

- Literature:

Introduction Techniques for 3D-Computer Vision
Emanuelle Trucco und Alessandro Verri, Prentice-Hall

Computer Vision: A Modern Approach
David Forsyth, Jean Ponce

What is this class about?

- How to do measurements with error assessment that is necessary for robotics
- Understanding of the image formation – what happens on the way from the light source to the pixel on the computer
- Projective Geometry and Accuracy evaluation of image processing steps
- Shape-from-X
- Omnidirectional vision

What is Computer Vision?

- **Trucco and Verri**
 - *computing properties of the 3D world from one or more digital image*
- **Stockman and Shapiro**
 - *To make useful decisions about real physical objects and scenes based on sensed images*
- **Ballard and Brown**
 - *The construction of explicit, meaningful description of physical objects from images*

What Information is in Images?

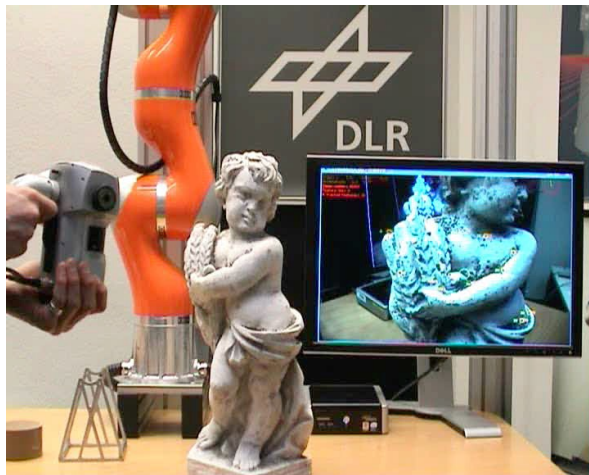


Vision, Darius Burschka

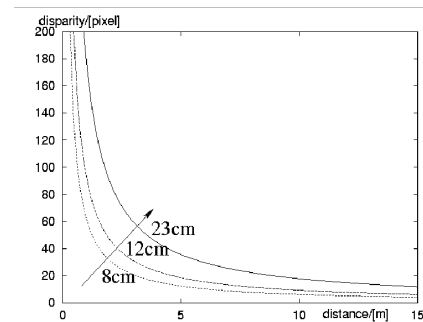
What Information is in Images?



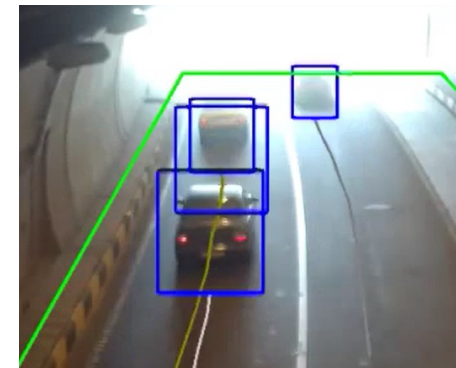
Problems with Camera-based Systems for Control



Camera by itself is too slow with 25-30Hz to stabilize a robot or monitor high dynamic motion

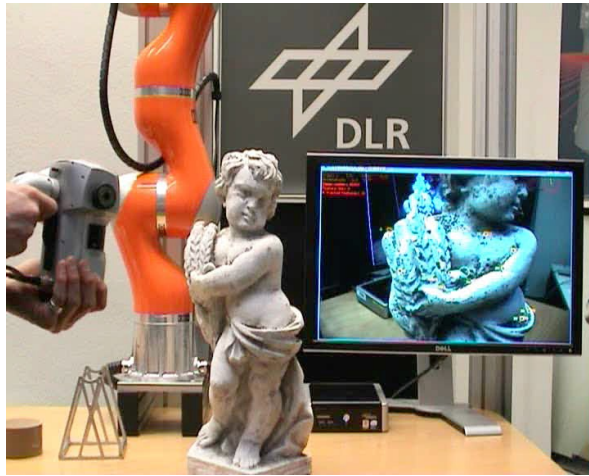


The quality of the reconstructed pose varies with the distance to the observed objects

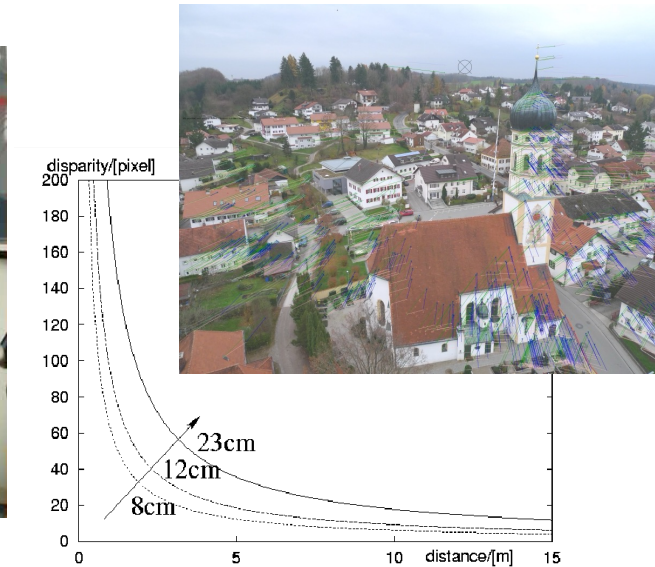


Camera can be blinded for multiple seconds in tunnels, etc.

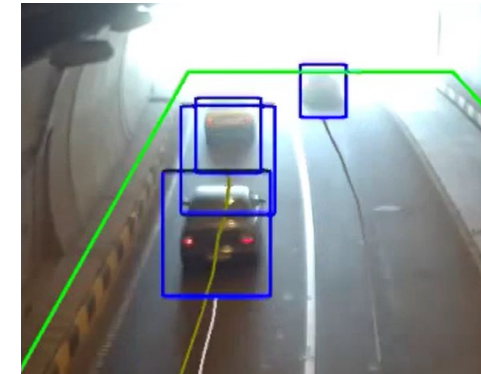
Problems with camera-based Measurement for Control



Camera by itself is too slow with 25-30Hz to stabilize a robot or monitor high dynamic motion



The quality of the reconstructed pose varies with the distance to the observed objects



Camera can be blinded for multiple seconds in tunnels, etc.

Pose Estimation (just vision)



Strobl, Mair, Bodenmüller,
Kielhofer, Sepp, Suppa, Burschka, Hirzinger
IROS, IEEE/RSJ, 2009, Best Paper Finalist
Mair, Strobl, Bodenmüller, Suppa, Burschka
KI, Springer Journal, 2010

Vision, Darius Burschka

Feature Propagation (Data fusion)

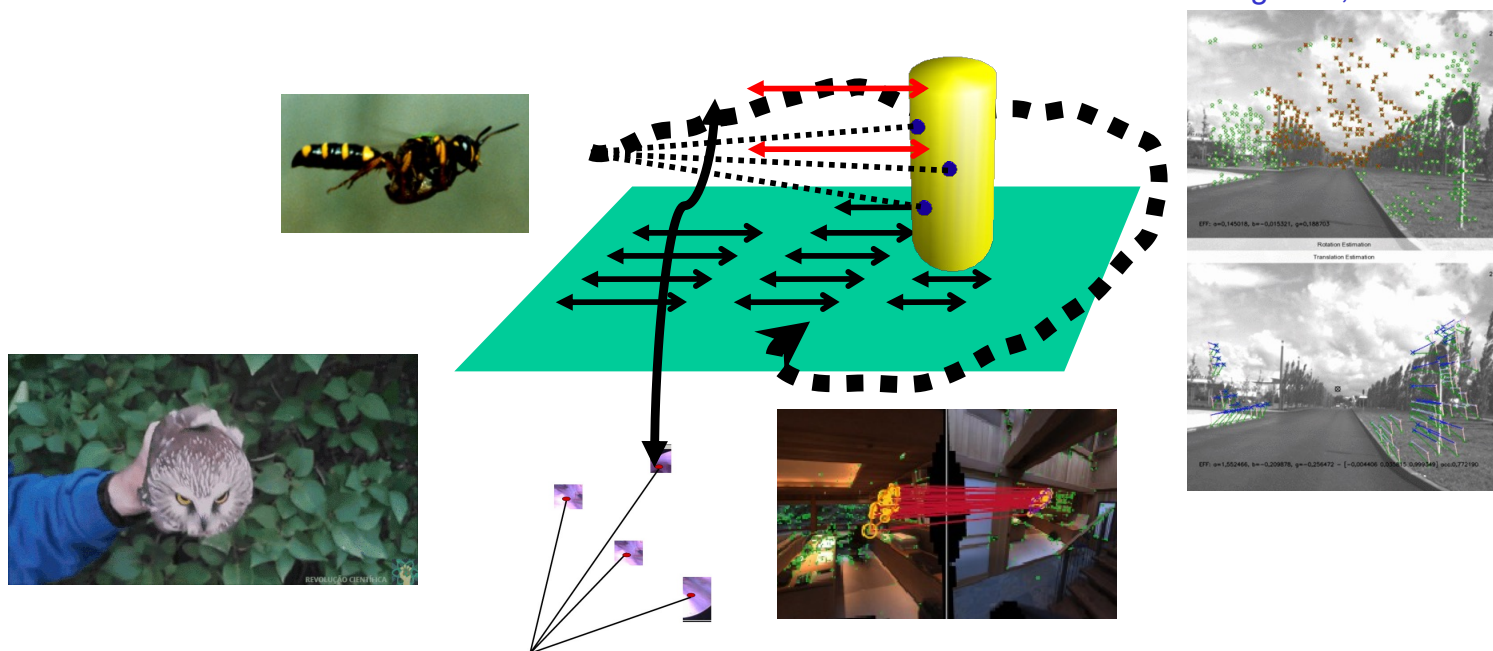


Strobl, Mair, Bodenmüller,
Kielhofer, Sepp, Suppa, Burschka, Hirzinger
IROS, IEEE/RSJ, 2009, Best Paper Finalist
Mair, Strobl, Bodenmüller, Suppa, Burschka
KI, Springer Journal, 2010

Vision, Darius Burschka

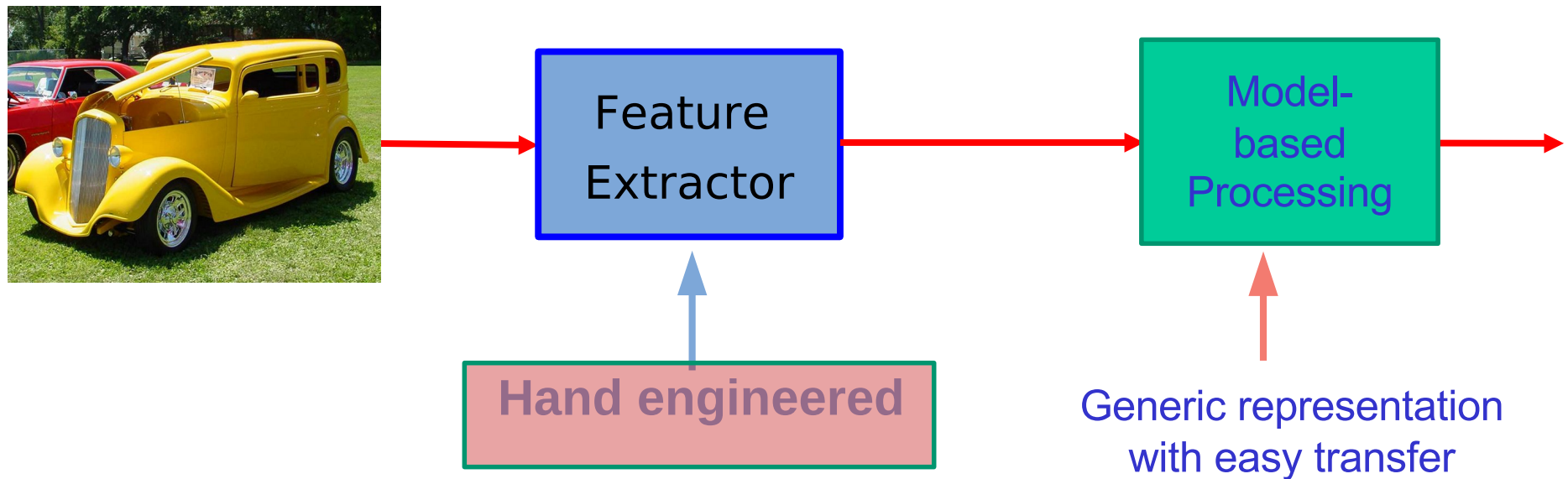
Biology helps with Data Fusion

Mair, Burschka
Mobile Robots Navigation, book chapter, In-Tech, 2010



Vision, Darius Burschka

Traditional Image Processing Framework



Current Trend to Avoid Handcrafted Features

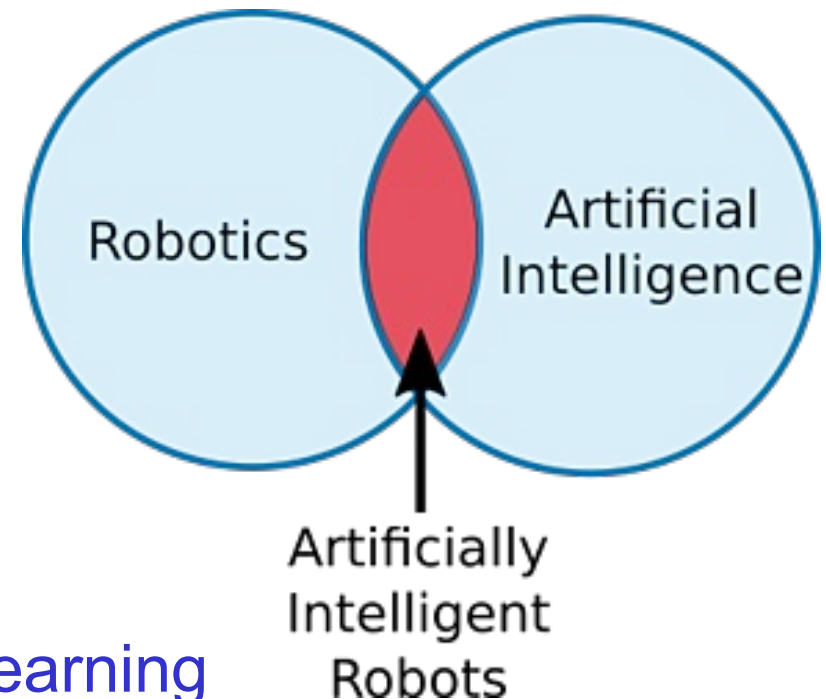
- Learning Approaches



- Semantics (object recognition) – ImageNet, VGG16
- Action modelling – RNNs

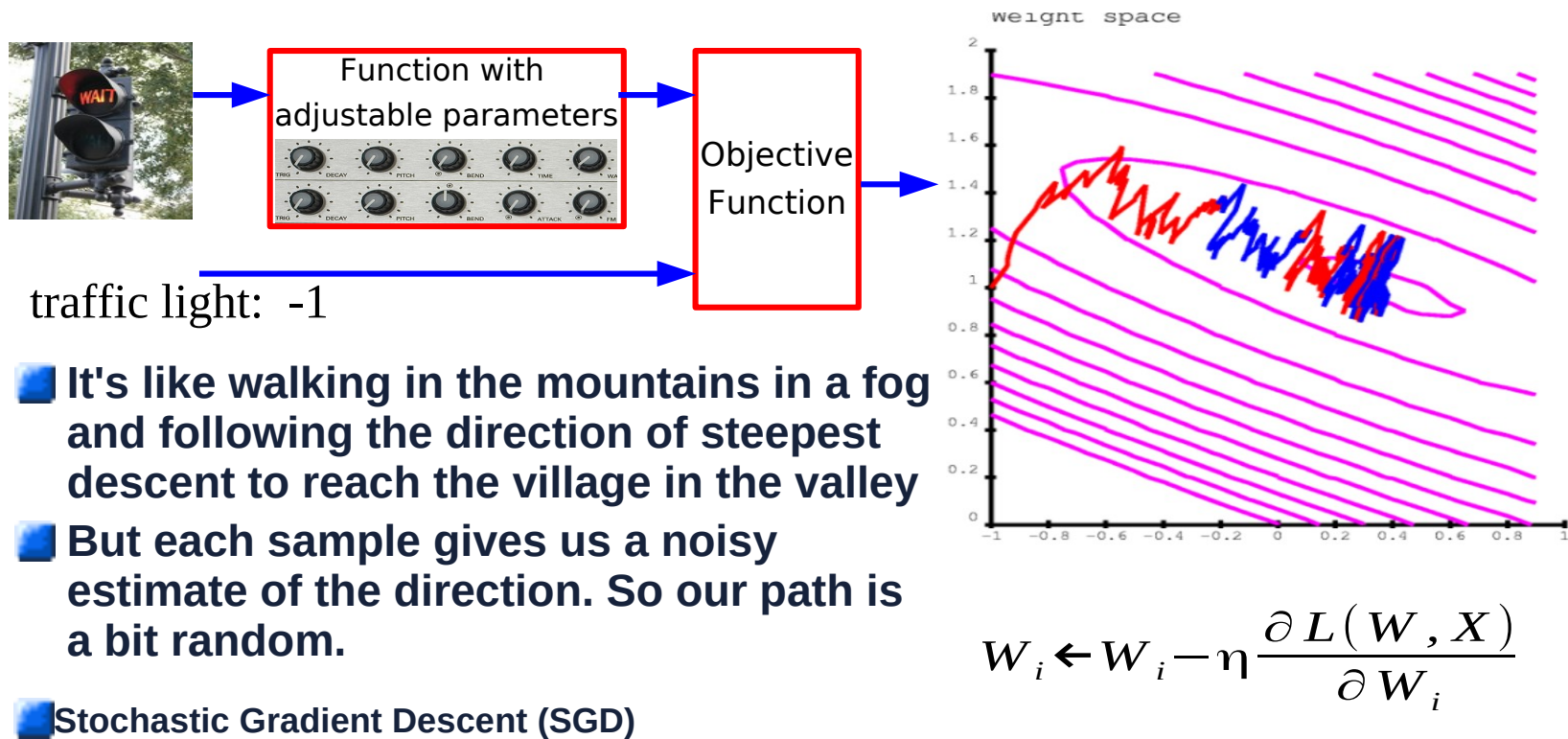
...

The machine learning system uses a large number of examples to learn the rules from observation.



Learning as Parameter Optimization without explicit Rules and Features

(Yann LeCun)



Emergence of Deep Learning – Big Data

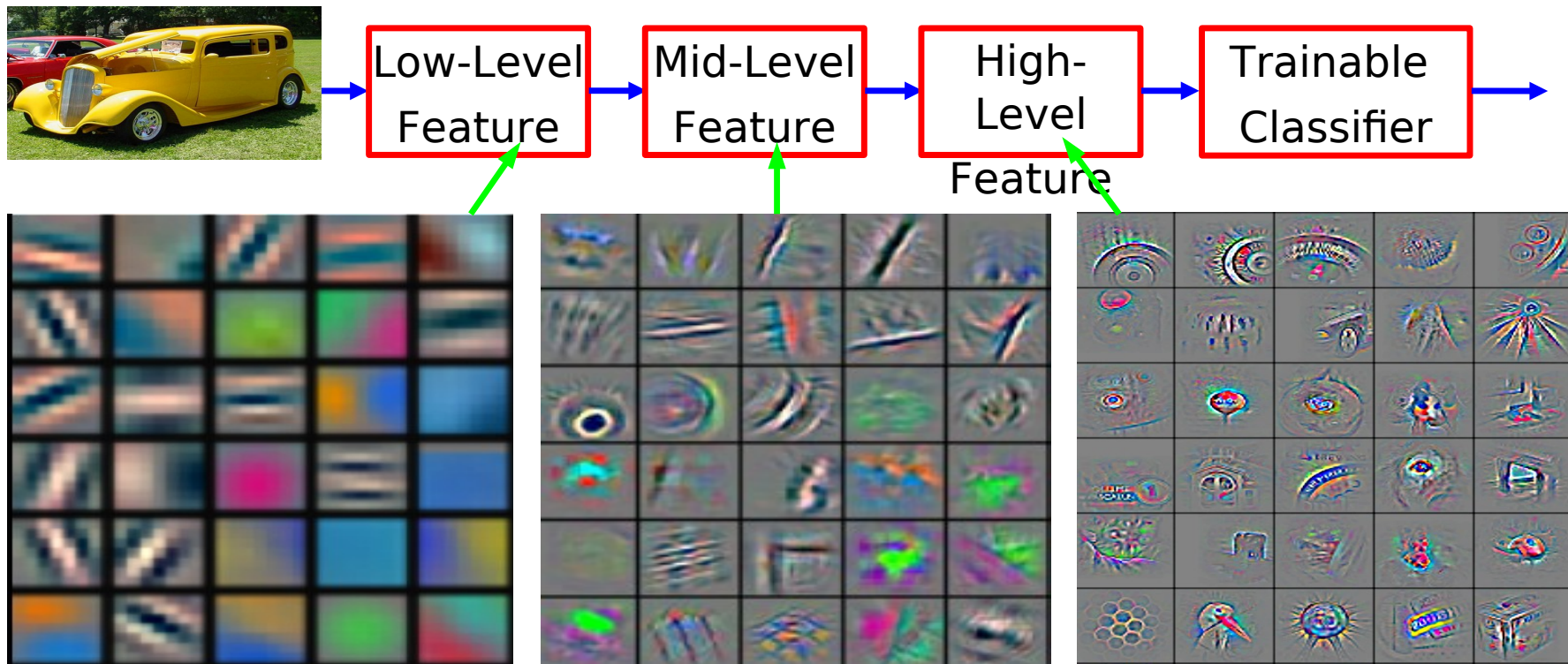


Textual or
visual query
to large
database
Labeling/Cat
egorization

Categorization of
behavioral models
for advertising and
news

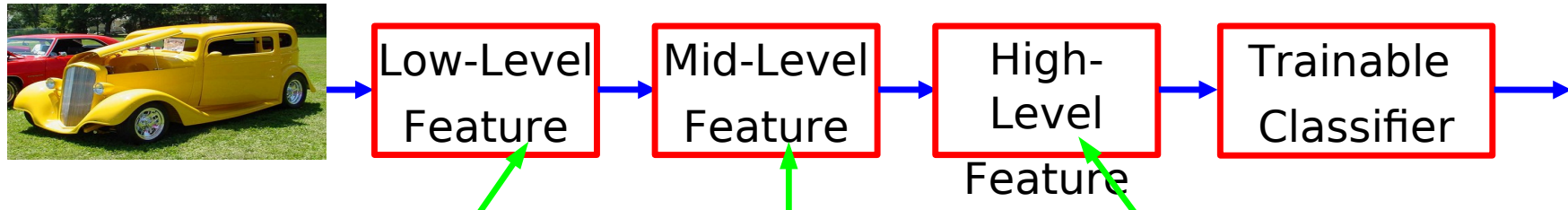
Similarity
measures and
shopping
behavior
modelling

Deep Learning = Learning Representations ?



Feature visualization of convolutional net trained on ImageNet from [Zeiler & Fergus 2013]

Explaining DL at CVision example is cheating...



Computer Vision is a well-studied and well-understood problem. This is what helps to formulate all the explanations but... you notice that it is hard to understand DL net structures on an unsolved problem.

Toy example to understand the problem:
DL framework that predicts partner's anger

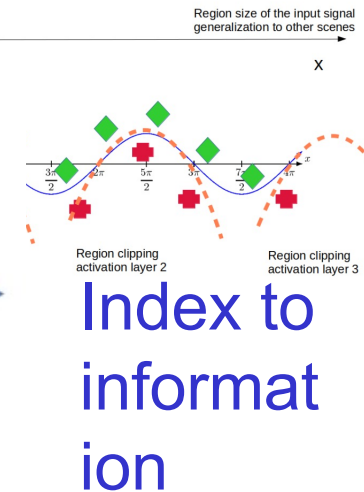
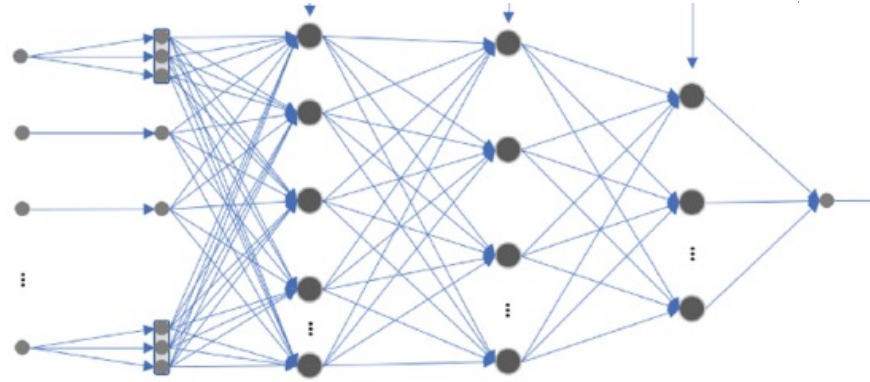


After a successful training, you will end up with the system that may correctly predict the anger state, but it will not help you understand the process and involved cues in any sense (black box)

What does the network represent?



Brightness values as a vector

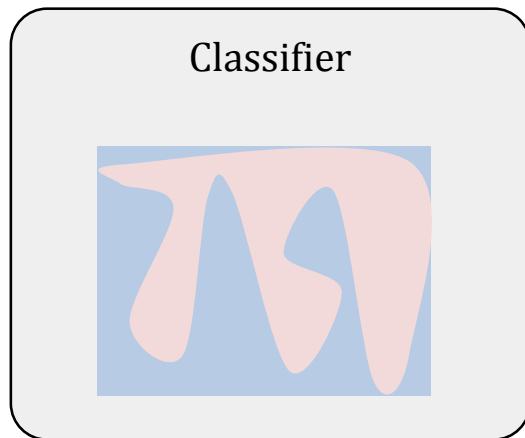


The network does not know, what it is trained on: structure, time of day or season? It matches an input vector to a label that can be used to access the information. Game with index numbers....

➡ DNNs are perfect to "pre-record" decisions resulting from complex model calculations to replay them in real-time during execution phase. Example: DeepMind controls Fusion Reactor with DNNs

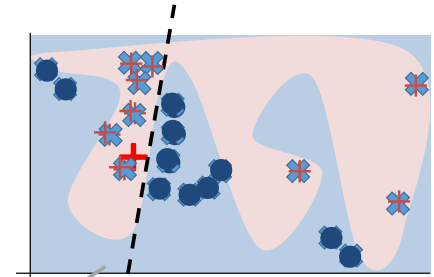
Can we understand the black box? → LIME

Global explanation may be too difficult



LIME: Sparse Linear Explanations

1. Sample points around x_i
2. Use model to predict labels for each sample
3. Weigh samples according to distance to x_i
4. Learn simple model on weighted samples
5. Use simple model to explain



LIME Example - Images (Certainty)

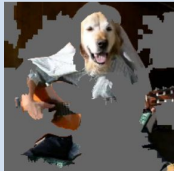
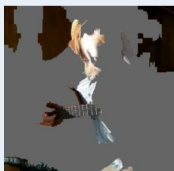


Original Image

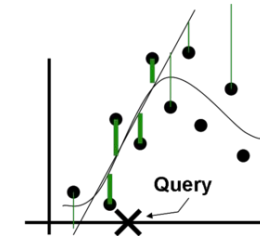
$P(\text{labrador}) = 0.21$

LIME is quite customizable:

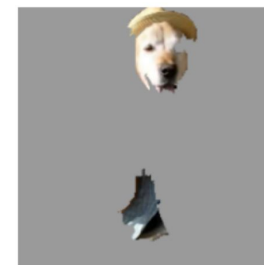
- How to perturb?
- Distance/similarity?
- How *local* you want it to be?
- How to express explanation

Perturbed Instances	$P(\text{Labrador})$
	0.92
	0.001
	0.34

Maybe to a fault?



Locally weighted regression



Explanation

Categorize Wolf vs Husky

[Ribeiro et al

Prediction wolf vs.
husky

Only 1 mistake!



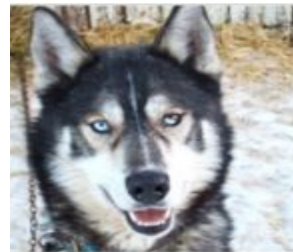
Predicted: **wolf**
True: **wolf**



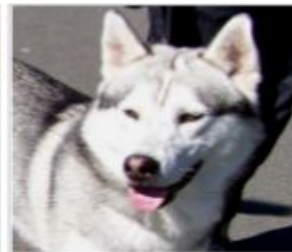
Predicted: **husky**
True: **husky**



Predicted: **wolf**
True: **wolf**



Predicted: **wolf**
True: **husky**



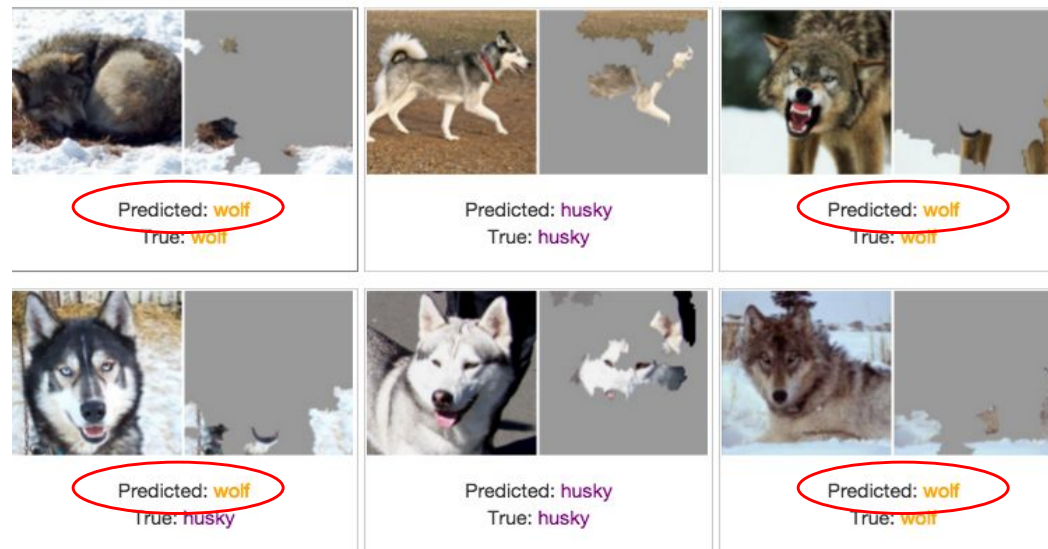
Predicted: **husky**
True: **husky**



Predicted: **wolf**
True: **wolf**

System learned the wrong classifier because of not enough diversity in the training set

[Ribeiro et al. 2016]



We've built a great snow detector...

25

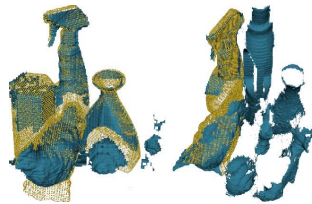
Our Research

Perception for manipulation

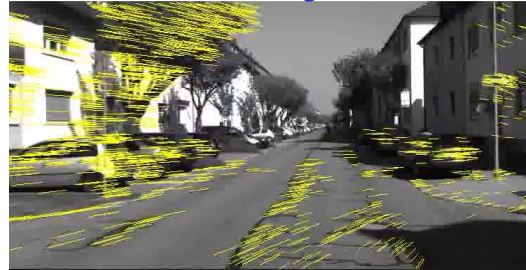


Rigid and Deformable
Registration

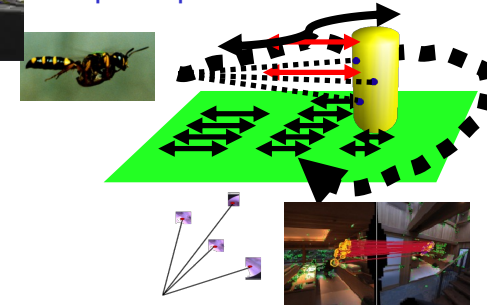
Rigid and Deformable
Registration



Visual navigation



Biologically motivated
perception

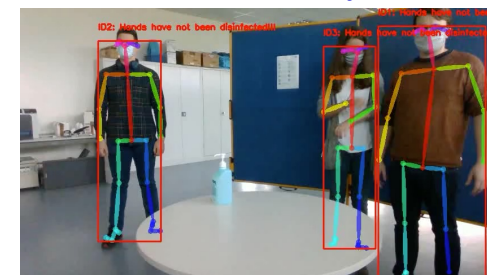


The Machine Vision and
Perception Group @TUM works
on the aspects of visual
perception and control in
medical, mobile, and HCI
applications

Photogrammetric monocular
reconstruction



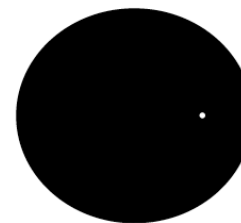
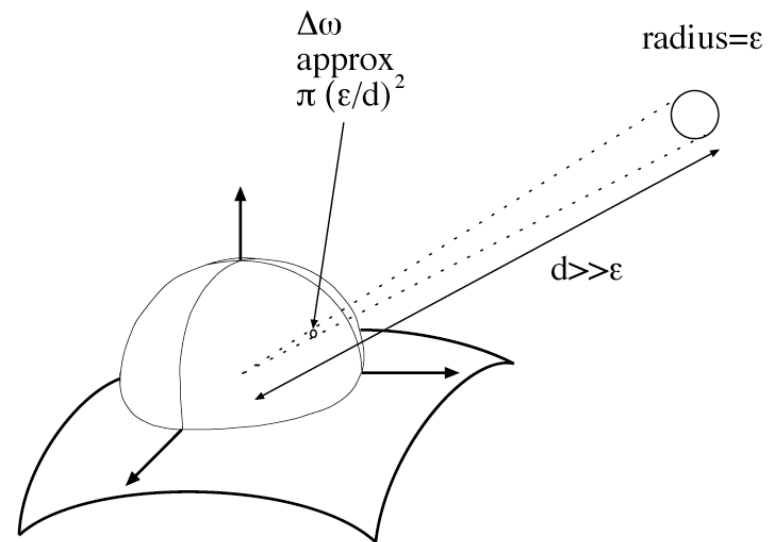
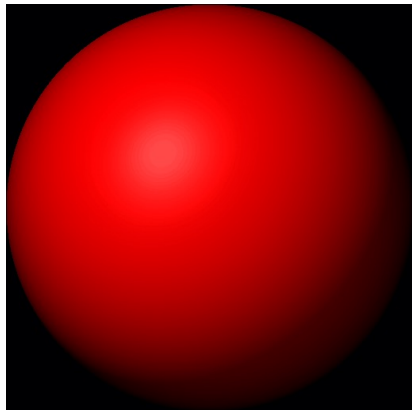
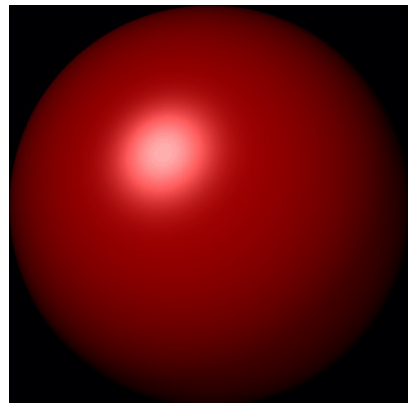
Visual Action Analysis





Problems of Computer Vision: Modeling

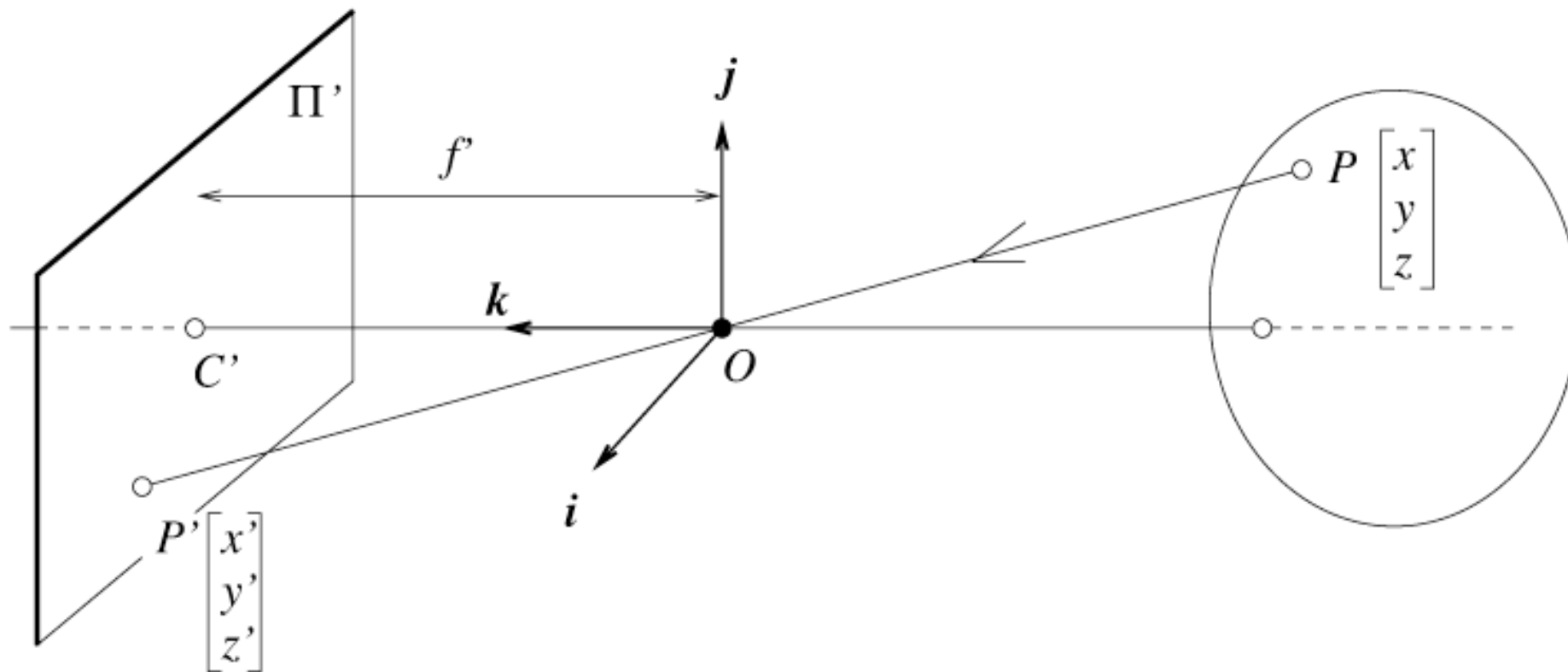
What are the physical and geometric processes that govern (digital) imaging?



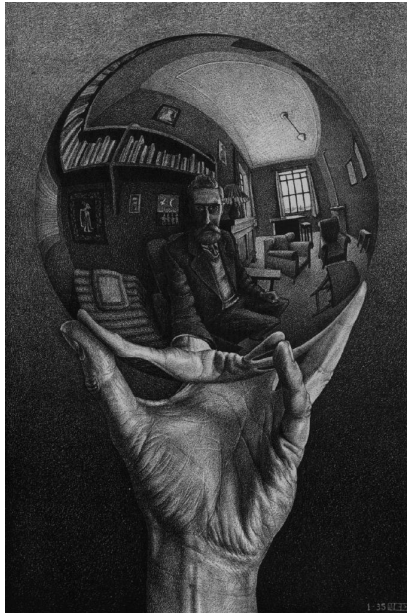
Constant
radiance patch
due to source

Problems of Computer Vision: Modeling

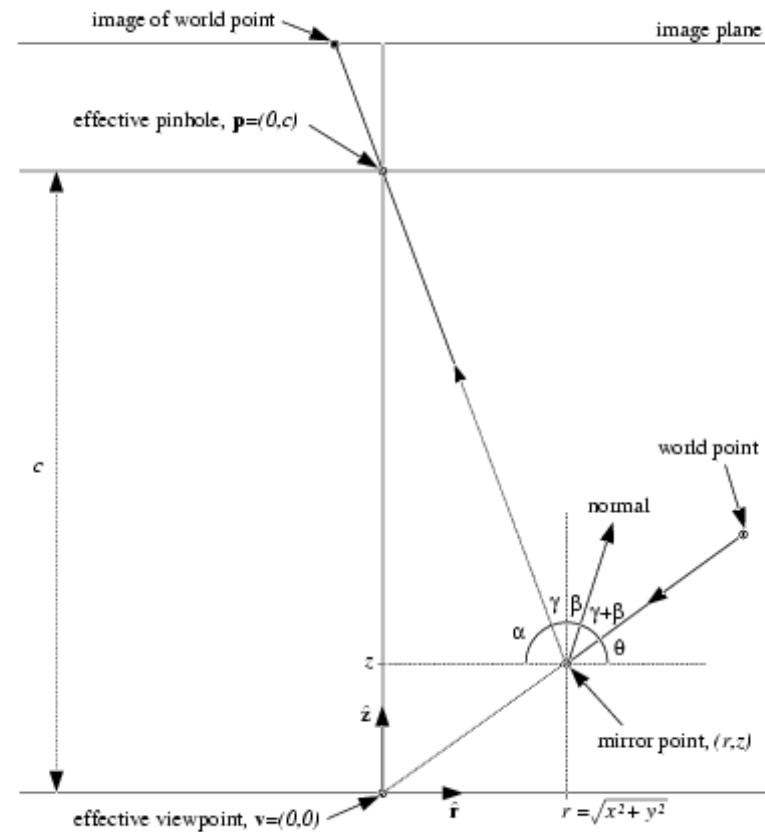
What are the physical and geometric processes that govern (digital) imaging?



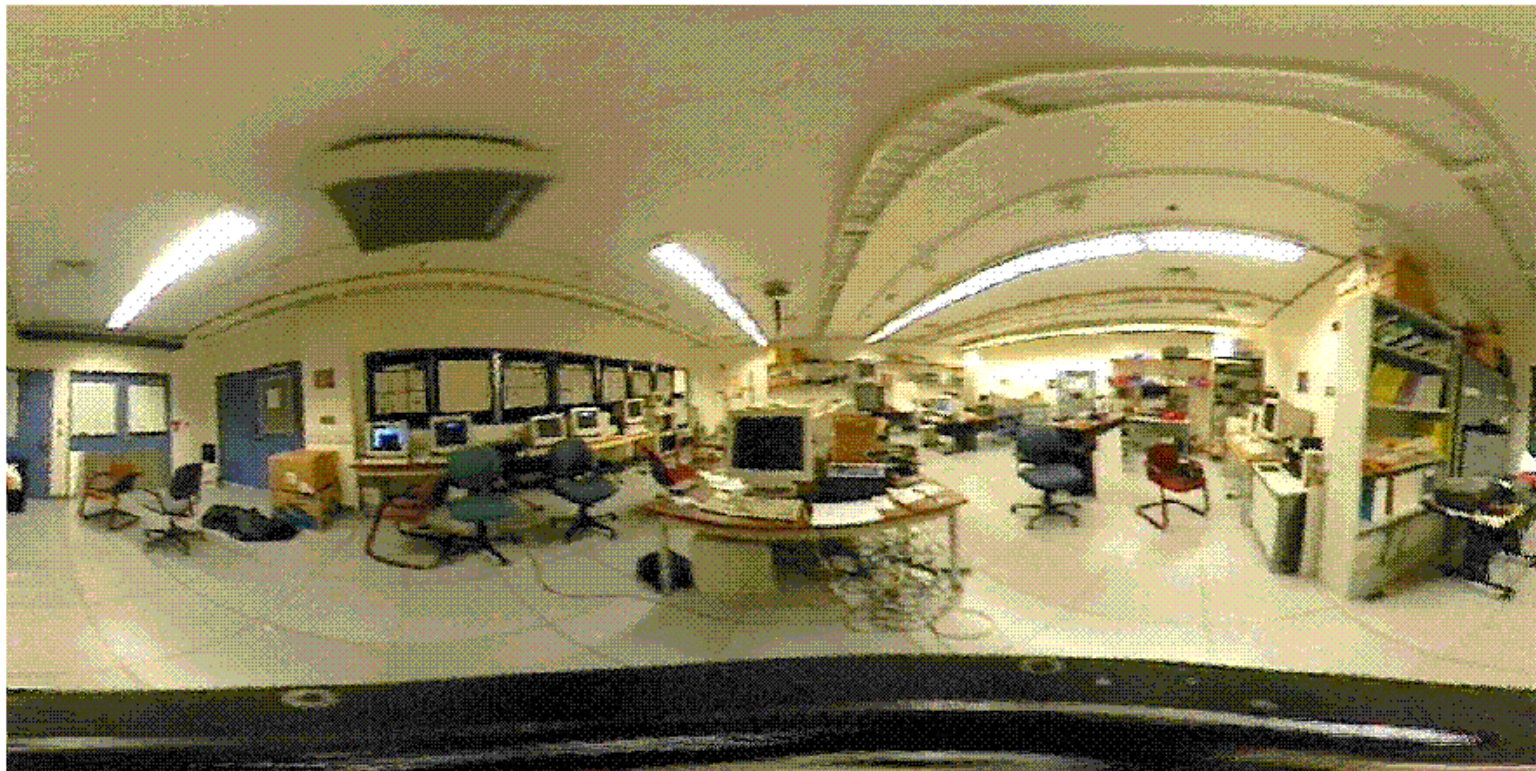
Omnidirectional Vision



Escher: *Hand with
Reflecting
Globe*



Examples



Computer Vision vs. Graphics

Is Vision the “Inverse” of Animation

- Computer Graphics
 - Produce “plausible” images
 - You choose the models, conditions, imaging parameters, etc.
- Computer Vision
 - Given real images with noise, sampling artifacts ...
 - Estimate physically quantities
 - Ill-posed ---- what is the minimum world knowledge we need?

Problems of Computer Vision: Feature Extraction

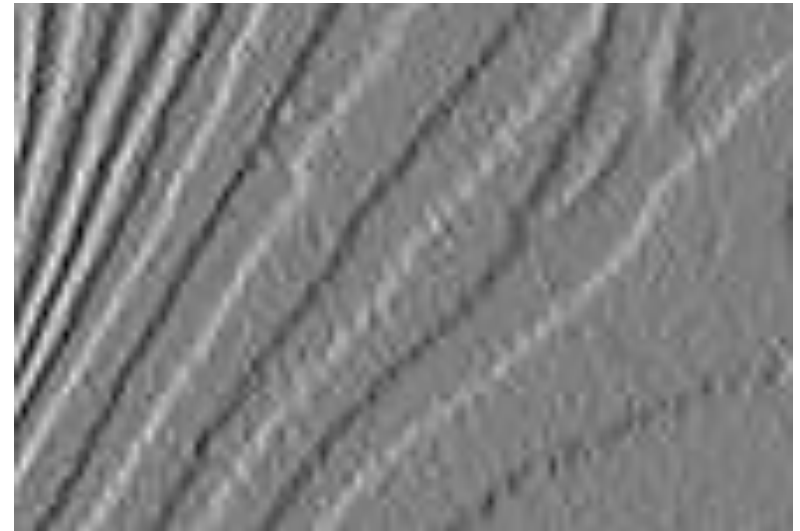
What are the “informative” areas of an image and
how do we detect them?



Image

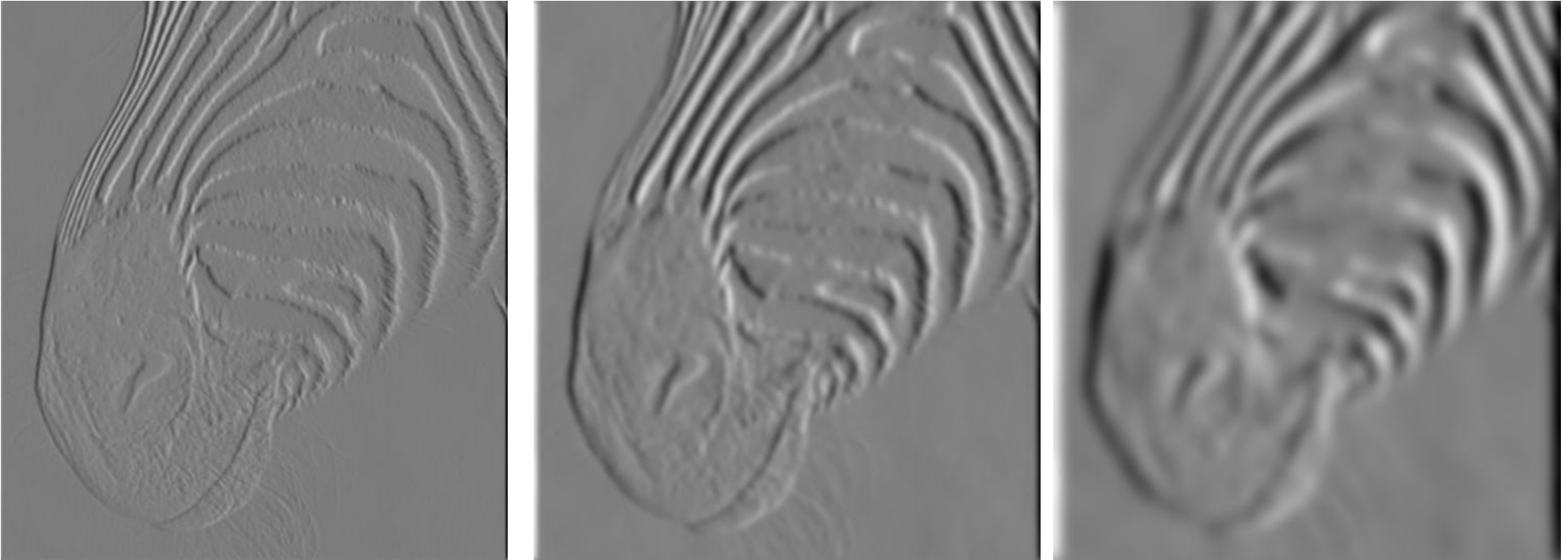


Filter



Result

Problems of Computer Vision: Feature Extraction



Filter kernels that are larger see effects at coarser scales -- the filter on the left responds to the zebra's whiskers, that on the right to its stripes

Problems of Computer Vision: Feature Extraction

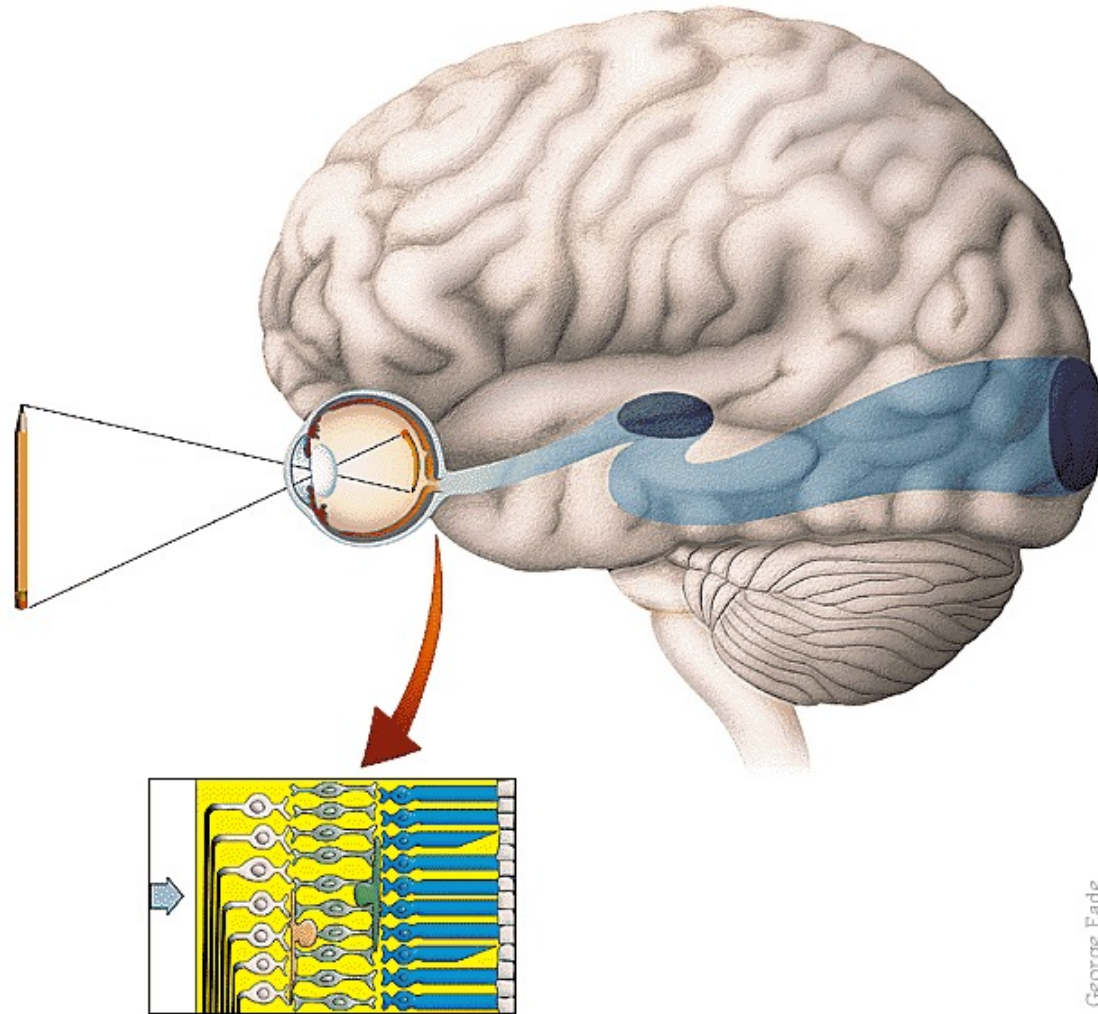


Thresholding suppresses “non-feature” areas
of the image

Computer Vision vs. Image Processing

- Image Processing
 - Mostly concerned with *image-to-image* transformations
 - Filtering
 - Enhancement
 - Compression
- Computer Vision
 - Concerned with how images *reflect the 3D world*
 - Filtering *for feature extraction*
 - Enhancement *for recognition/detection*
 - Compression *that preserves geometric information in images*

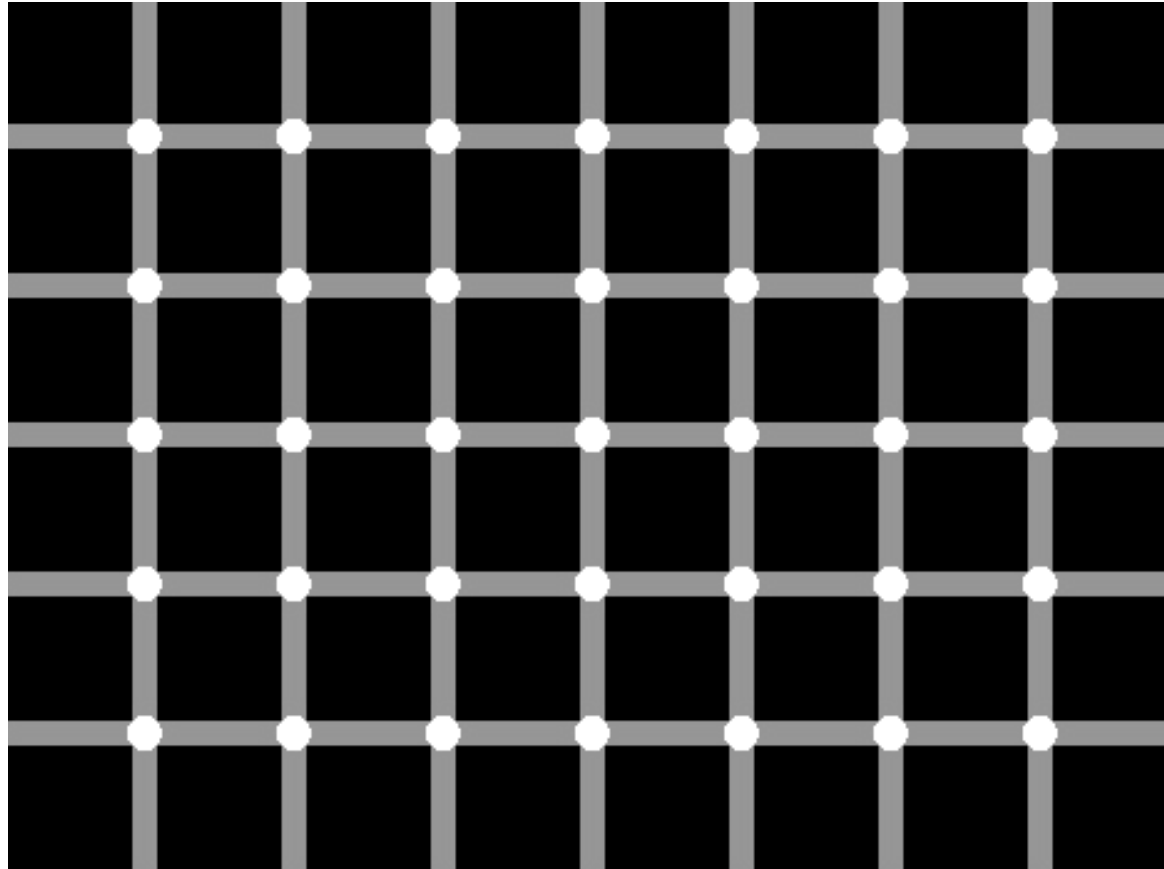
Computer Vision vs. Human Vision



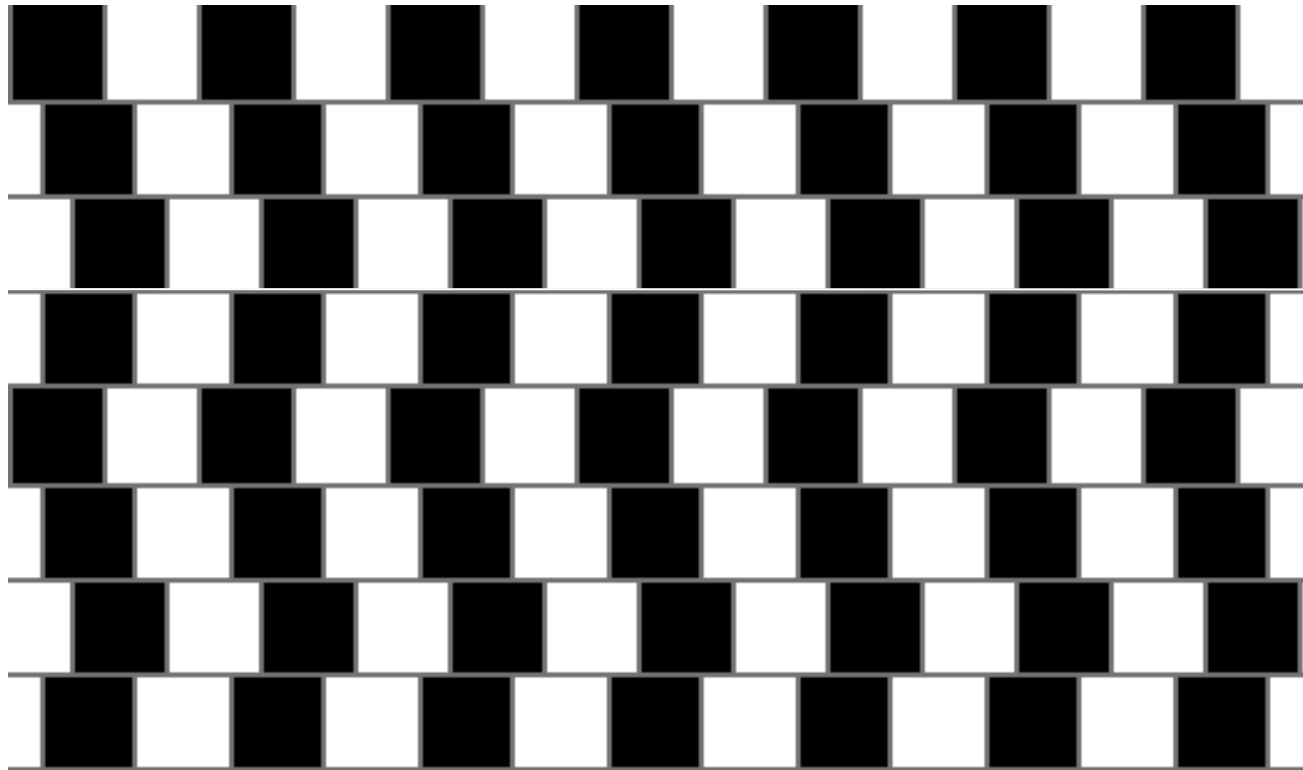
What is the right segmentation?
To us it seems obvious ...

George Eade

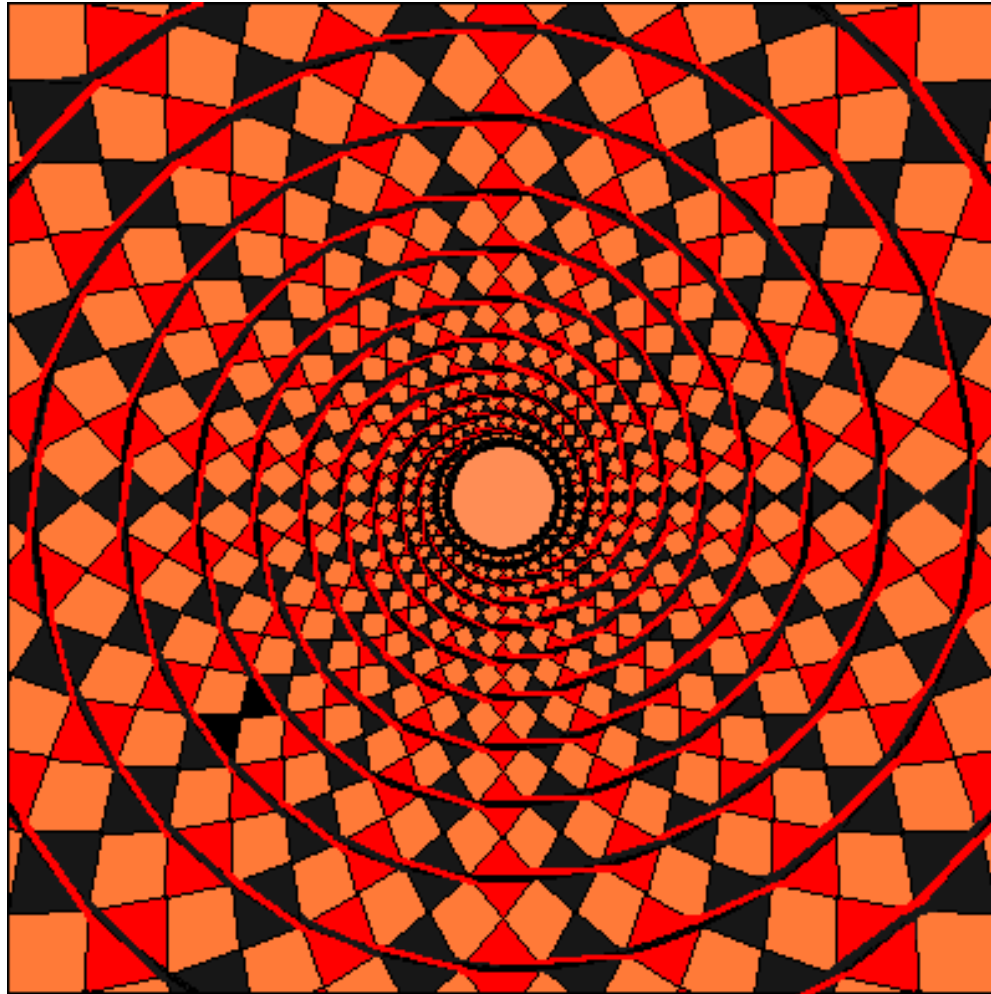
Illusions: What Do They Tell Us?



Illusions: What Do They Tell Us?



Illusions: What Do They Tell Us?



Illusions: What Do They Tell Us?

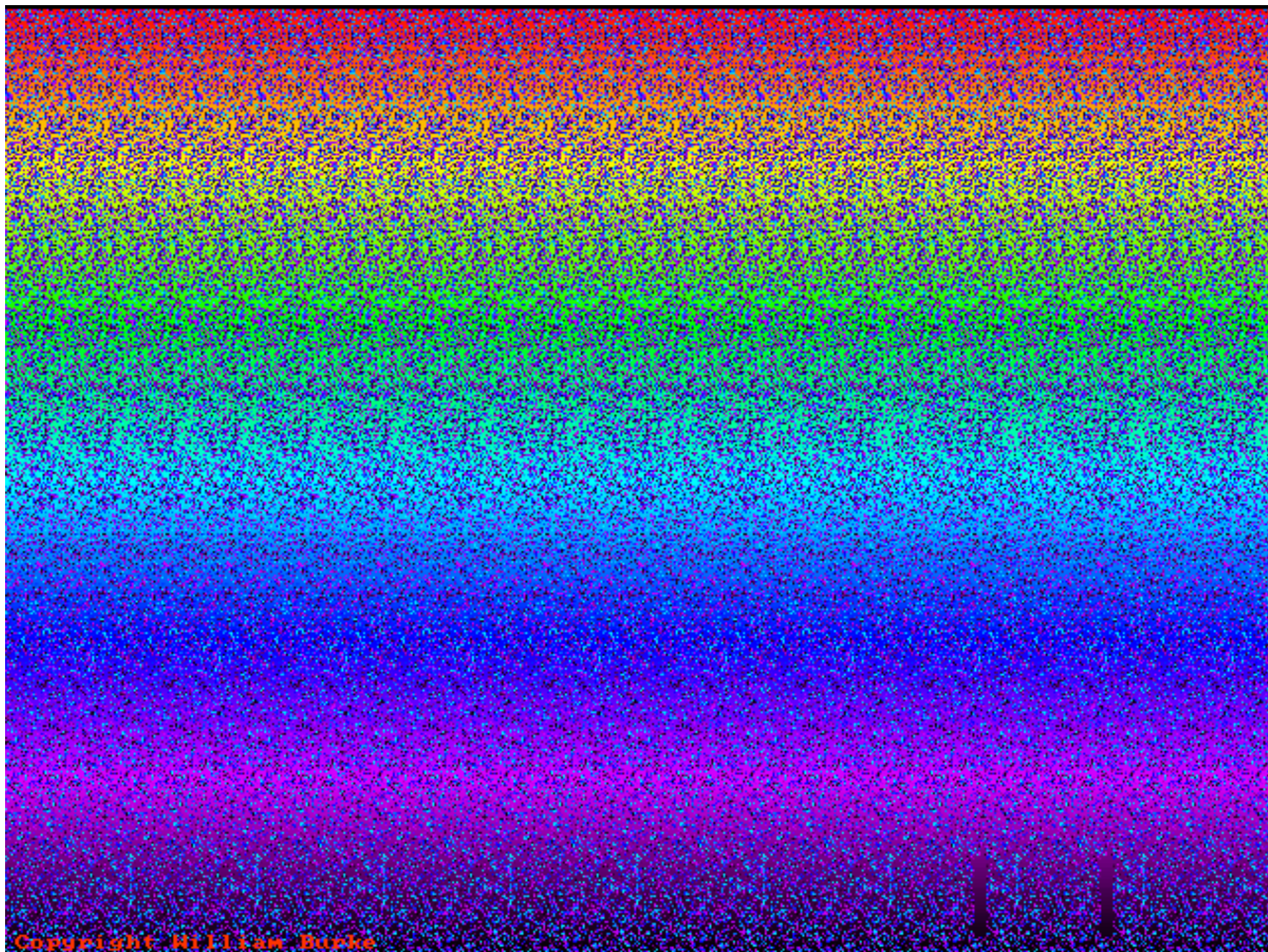


Vision, Darius Burschka

Illusions: What Do They Tell Us?



Vision, Darius Burschka



Problems of Computer Vision: Motion Estimation

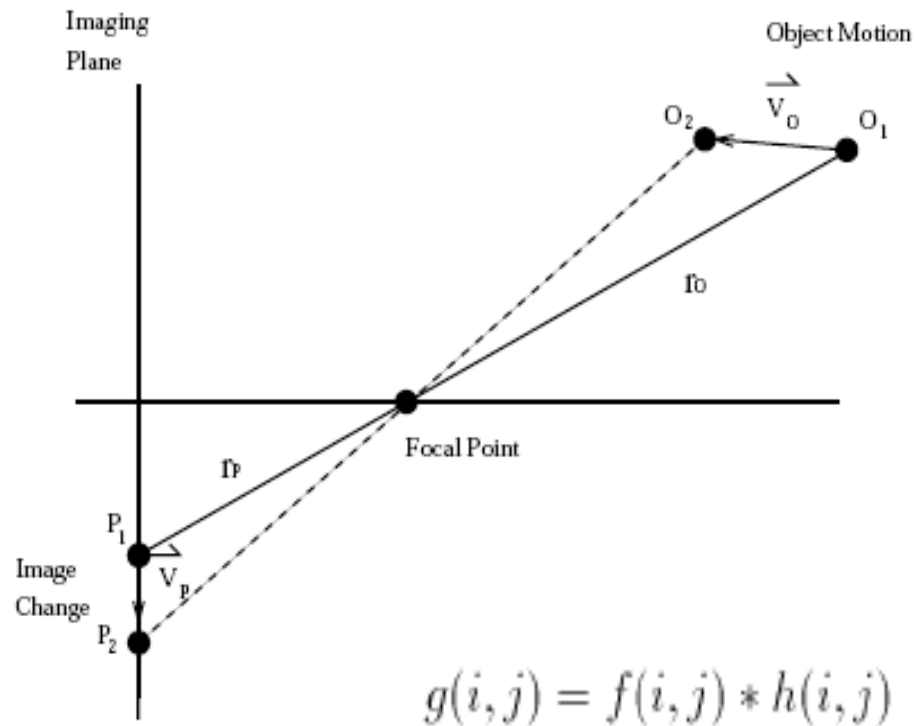
Detecting motion:



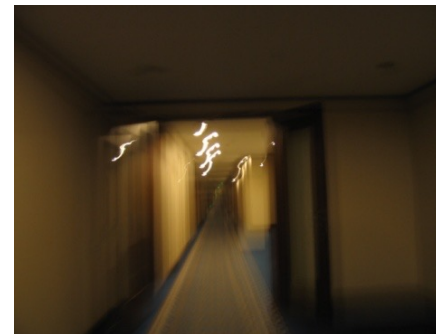
Problems of Computer Vision: Motion Estimation



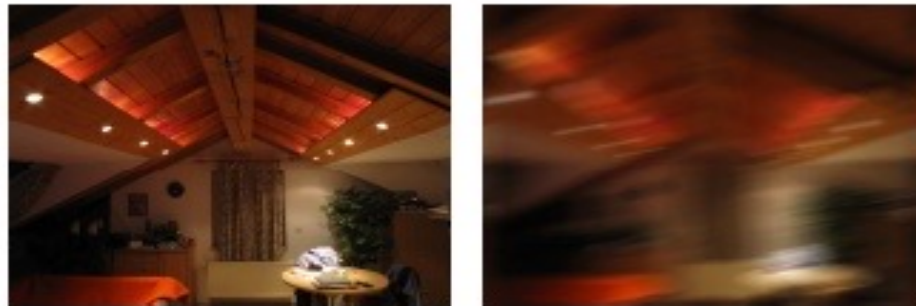
Point Spread Function (PSF)



$$h(x, y) = \begin{cases} \frac{1}{d}, & 0 \leq |x| \leq d \cdot \cos \alpha \wedge y = \sin \alpha \cdot x \\ 0, & \text{otherwise} \end{cases}$$



Motion Blur to Support Tracking



$$G(u, v) = I(u, v) \cdot H(u, v),$$

$$h(x, y) = \begin{cases} \frac{1}{d}, & 0 \leq |x| \leq d \cdot \cos \alpha \wedge y = \sin \alpha \cdot x \\ 0, & \text{otherwise} \end{cases}$$

$$F(u, v) = \mathcal{F}\{f(x, y)\} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) e^{-i(ux+vy)} dx dy$$

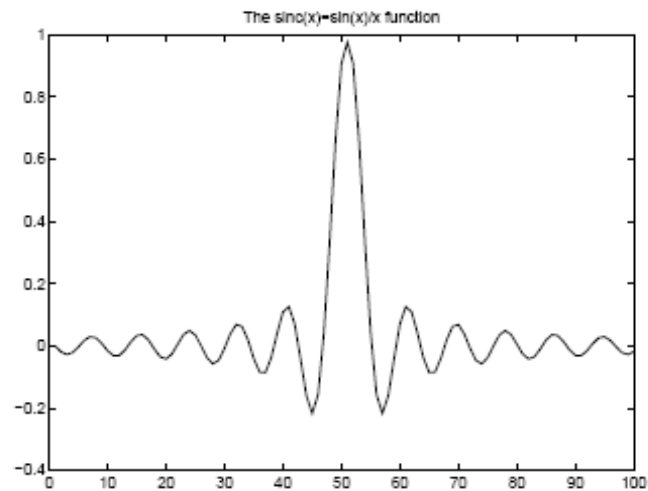
$$f(x, y) = \mathcal{F}^{-1}\{F(u, v)\} = \frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(u, v) e^{i(ux+vy)} du dv$$

$$H(u, v) = \frac{\sin(\pi du)}{\pi du} = \text{sinc}(\pi du)$$

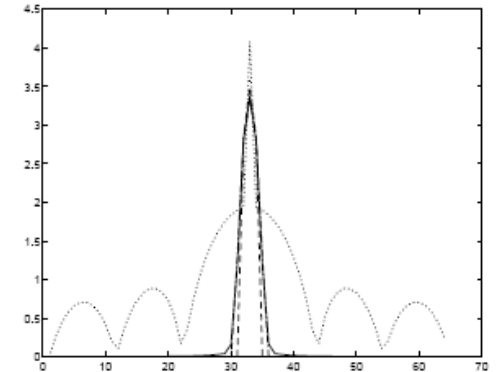
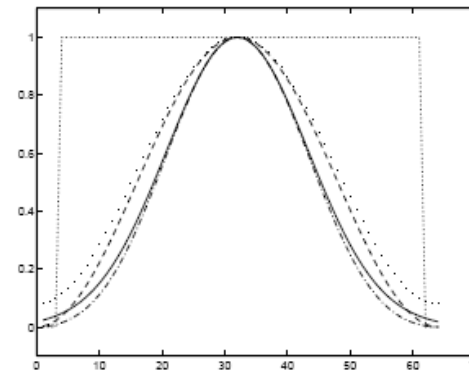
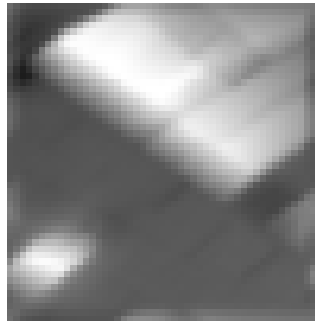
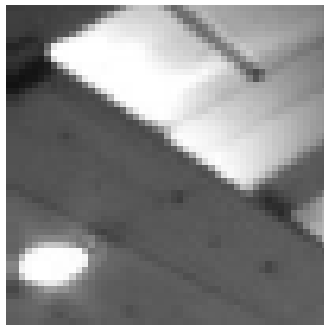
Cepstrum

The Cepstrum is the Fourier transformation of the log spectrum of an image it is therefore a tool for analyzing the frequency domain of an image

$$C_f(p,q) = \mathcal{F}^{-1} \{ \log |F(u,v)| \}$$



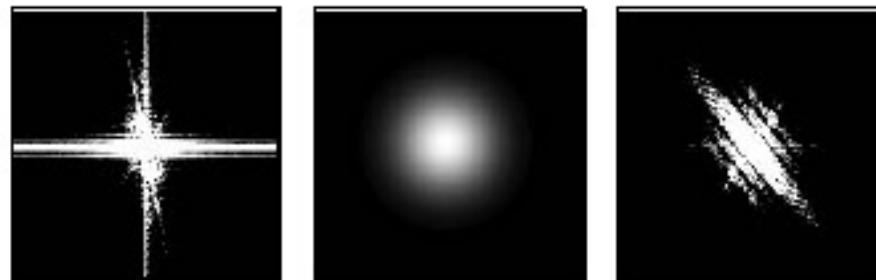
Smoothing to Reduce Window Artifacts



Magnified image region

Artifacts

Gaussian window to avoid artifacts in Cepstrum



Real-Time Navigation Data from an Image Sequence



Vision, Darius Burschka

Estimation of the 6 Degrees of Freedom



We used to reconstruct static scenes from monocular in
2007... (DLR)



Accuracy: 1.5cm

Vision, Darius Burschka

Applications of Computer Vision: Biometrics

- Face recognition
- Iris scanning
- Fingerprint recognition
- Activity recognition

Applications of Computer Vision: Medical Imaging

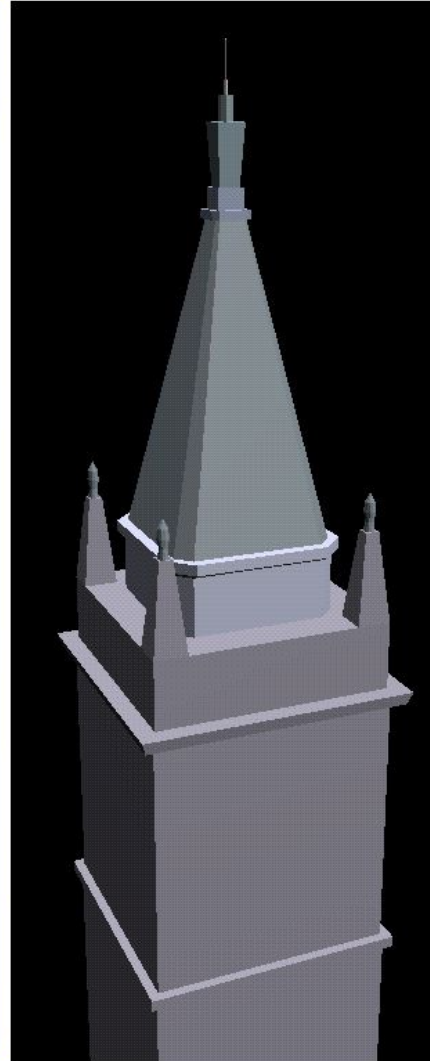


Applications of Computer Vision: Image Databases

(Courtesy D. Forsyth & J. Ponce)



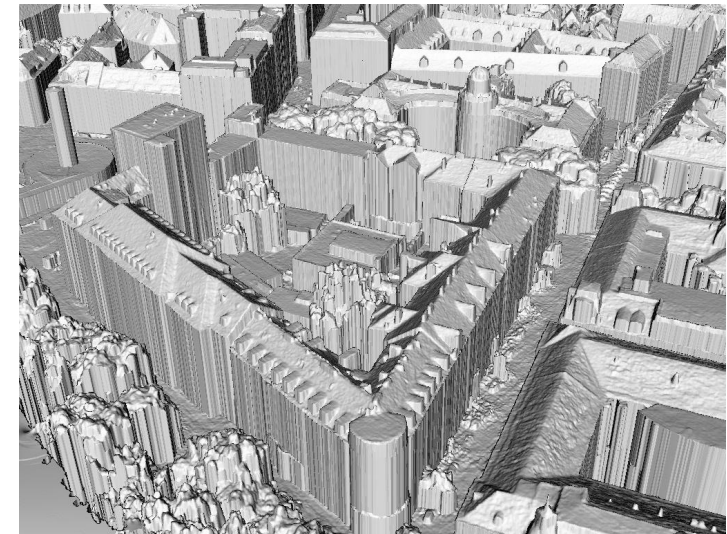
Applications of Computer Vision: Data Acquisition



Geometry-based environment modeling



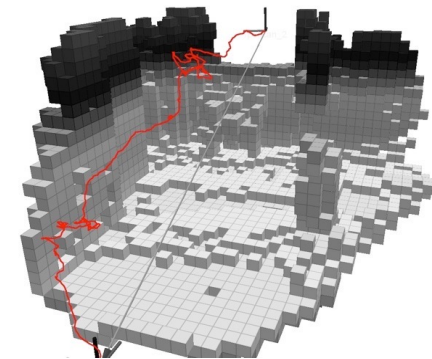
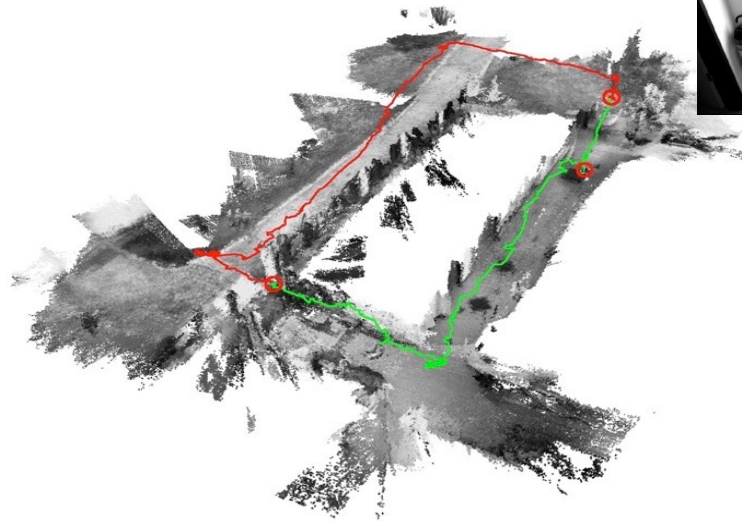
Geometrical Reconstruction from Aerial-Vehicles



High spatial accuracy of the geometrical reconstruction, but limited volumetric resolution

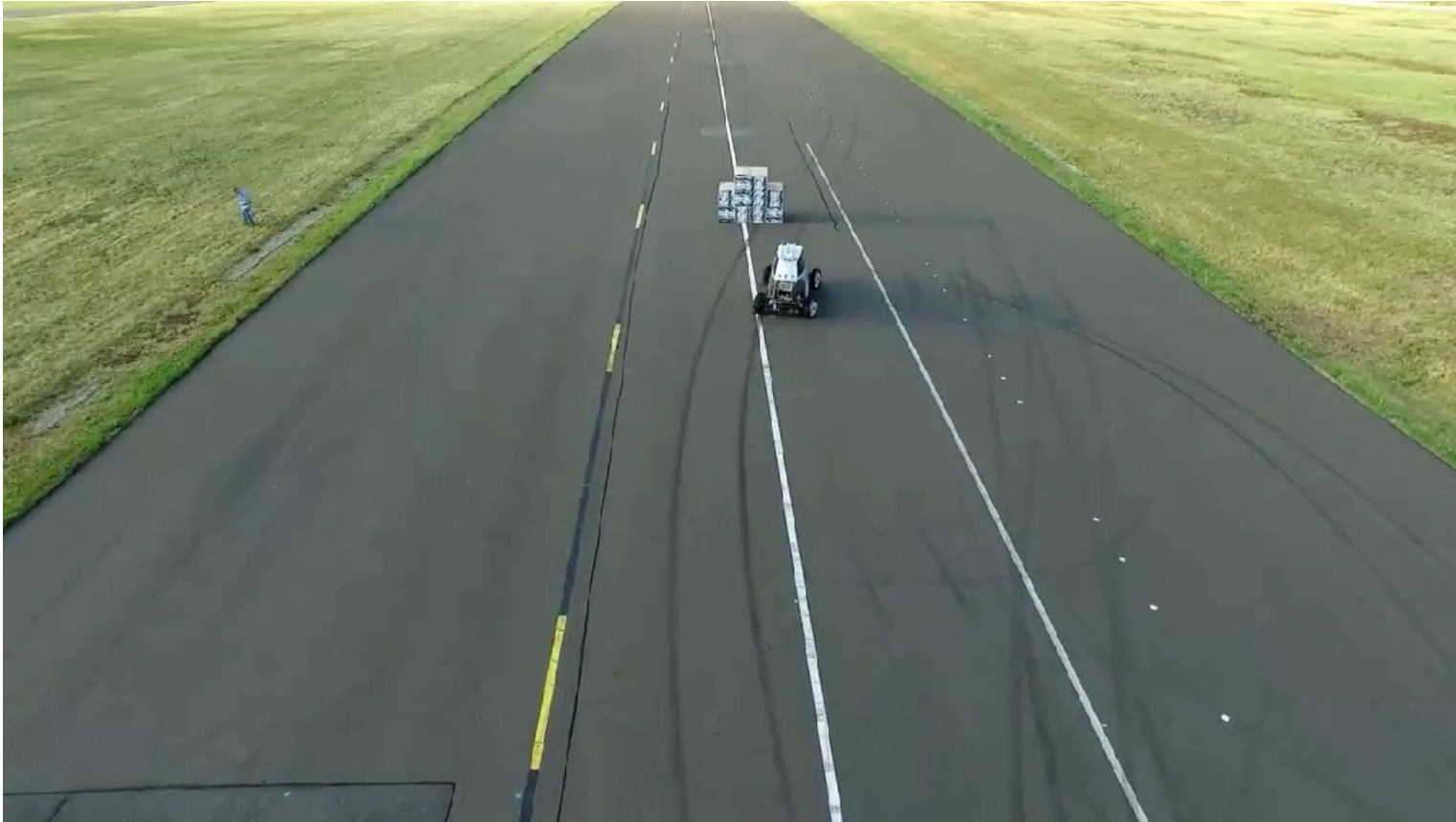
Navigation under strong illumination changes

- Autonomous indoor/outdoor flight of 60m
- Mapping resolution: 0.1m
- Leaving through a window
- Returning through door



Vision, Darius Burschka

Obstacle Avoidance in Dynamic Spaces



Course Information

- Use the course WEB site
 - <https://mvp.in.tum.de/courses/vision/>
- For information on this course